

OL P1 -MOCK 2025 - MS

Computer Science 2210/0478



Zak
ZAFAR ALI KHAN



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Zak
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Student Name:	
Student Cell Phone (WhatsApp):	
Current Centre:	
Registered in Branch (Jauhar/Gulshan/North/Bahadurabad):	



Paper 1

1 A student is using a smartphone.

Part	Answer	Marks
(a)	Any two input devices built into a smartphone:	1 mark each (Max 2)
	• Touchscreen (as an input device)	
	• Microphone	
	• Camera	
	• Accelerometer / gyroscope / fingerprint scanner	
(b)	Any one output device built into a smartphone:	1 mark
	• Screen (display)	
	• Speaker	
(c)	Any one storage device built into a smartphone:	1 mark
	• Solid State Drive (SSD) / Flash memory / Internal memory	

2 Hypertext markup language (HTML) colour codes can be represented as hexadecimal.

(a) Tick (✓) **three** boxes to show which statements about the hexadecimal number system are correct.

A It uses the values 0 to 9 and A to F.



B It can be used as a shorter representation of binary.



C It is a base 10 system.



D It can be used to represent error codes.





Paper 1

(b) Denary numbers can be converted to hexadecimal.

Convert the **three** denary numbers to hexadecimal.

Denary	Hexadecimal	Marks
20	14	1
32	20	1
165	A5	1

[3]

3 The binary number 10100011 is stored in random access memory (RAM).

A logical left shift of **three** places is performed on the binary number.

(a) Give the 8-bit binary number that will be stored after the shift has taken place.

Original: 1 0 1 0 0 0 1 1

Shifted: 1 1 0 0 0 0 0 0

[1]

(b) Tick (✓) **one** box to show which statement about a logical left shift of **two** places is correct.

A It would divide the binary number by 2.

☐

B It would multiply the binary number by 2.

☐

C It would divide the binary number by 4.

☐

D It would multiply the binary number by 4.

☒

(c) 10100011 can be stored as a two's complement integer.

Convert the two's complement integer 10100011 to denary. Show all your working.

-93

[2]

(d) The binary number is measured as a byte because it has 8 bits.

State how many bytes there are in a kibibyte (KiB).

1024

[1]





Paper 1

4 A school sends student data from one computer to another using a network.

(a) What is meant by the term data packet?

Acceptable points (**Any two** for full marks):

-  A unit of data for transmission
-  Contains part of the overall data/message
-  Includes control information (e.g., destination address, source address, packet number, error-checking data)
-  Sent individually over the network

[2]

(b) The data is sent using packet switching.

Explain how packet switching works.

Acceptable points (**Any three** for full marks):

-  Data is split into packets
-  Each packet is sent independently across the network
-  Packets can take different routes to the destination
-  Packets are reassembled in the correct order at the destination
-  Missing/corrupt packets are detected and re-sent

[3]

(c) The data is sent using serial transmission.

Give two reasons why serial transmission is used instead of parallel transmission.

Acceptable points (**Any two** for full marks):

-  Serial transmission is more reliable over longer distances
-  No risk of signal skew (bits arriving at different times)
-  Less hardware/wiring needed (cheaper and simpler)
-  More suitable for transferring data through cables like USB

[2]

5 A tablet computer uses memory and storage to work with programs and data.

(a) What is the purpose of **RAM** in a computer?

Acceptable points (**Any two** for full marks):

-  RUNs the computer.
-  RAM stores data, files, and programs that are currently in use.
-  RAM is temporary/volatile memory (data is lost when power is off).

[2]





Paper 1

(b) The tablet also uses **secondary storage**.

Explain why secondary storage is needed in a computer.

Acceptable points (**Any two** for full marks):

-  To permanently store programs and data (non-volatile storage).
-  Retains data when the computer is turned off.
-  Provides large capacity storage compared to RAM.

[2]

(c) Tick (✓) **three** devices that are examples of **secondary storage**.

- ☐ Random Access Memory (RAM)
- ☒ Compact Disk (CD)
- ☒ Blu-ray Disk
- ☐ Read Only Memory (ROM)
- ☒ Hard Disk Drive (HDD)
- ☐ Graphics Card

[3]

(d) The processor uses the **fetch–decode–execute (FDE) cycle** to run programs.

Describe what happens in the **cycle**.

Acceptable points (**Any four** for full marks):

-  Fetch the next instruction from memory (RAM).
-  Transfer instruction to the control unit.
-  Decode the instruction to determine the action needed.
-  Execute the instruction (perform operation, e.g., calculation, data movement).
-  Repeat the cycle for the next instruction.

[4]

6 A computer needs firmware and system software to operate.

(a) State the purpose of firmware.

Acceptable points (**Any one** for full mark):

-  Firmware provides low-level control for the device's hardware.
-  Firmware contains essential programs needed for hardware to function.
-  Firmware is permanent software programmed into hardware devices.

[1]

(b) Give **one** example of firmware.

Acceptable examples (**any one**):

-  BIOS (Basic Input/Output System)
-  UEFI (Unified Extensible Firmware Interface)
-  Firmware on printers, routers, keyboards, smartphones, etc.
-  Embedded software in washing machines, smart TVs.

[1]





Paper 1

(c) Give **two** examples of system software.

Acceptable examples (**any two**):

-  Operating system (e.g., Windows, Linux, Android)
-  Utility programs (e.g., antivirus software, disk defragmenter)
-  Device drivers
-  Compiler
-  Linker
-  Interpreter

[2]

7 A company sends secure messages using encryption.

Complete the paragraph about **asymmetric encryption**.

Choose words from the list. Some words will not be used. Use each correct word only once.

asymmetric cipher text encrypted key plain text
private key public key decrypted certificate symmetric

The sender starts with some **plain text**.

This data is **encrypted** using the receiver's **public key**.

Only the receiver can turn it back into plain text by using their **private key**.

[4]

8 A factory uses an automated robot to move boxes from one conveyor belt to another.

(a) What does it mean when a robot is described as **automated**?

Acceptable points (**Any one** for full mark):

-  The robot can perform tasks without human intervention.
-  It operates automatically based on programmed instructions.
-  It makes decisions or actions independently based on sensor input.

[1]

(b) List **three characteristics** of an automated robot.

Acceptable characteristics (**Any three** for full marks):

-  Works without human input
-  Can sense its environment (uses sensors)
-  Can make simple decisions
-  Repeats tasks accurately and consistently
-  Operates continuously
-  Performs tasks faster than humans
-  Performs dangerous tasks

[3]





Paper 1

(c) The robot moves boxes and must stop when it reaches the end of the conveyor belt. It uses **sensors** and a **microprocessor** to detect the end.

Explain how the robot uses the sensors and microprocessor to perform this task.

Acceptable points:

-  Sensors detect physical conditions (e.g., distance, object presence, end of conveyor).
-  Sensors send signals/data to the microprocessor.
-  Microprocessor processes the input data.
-  Microprocessor compares input to pre-set values.
-  Microprocessor makes a decision based on sensor input.
-  Microprocessor sends signals to stop the robot.

[6]

(d) Give **two** advantages of using an automated robot to move boxes in a factory.

Acceptable points (**Any two** for full marks):

-  Robots can work continuously without breaks.
-  Robots can reduce human error.
-  Robots can perform repetitive tasks faster and more accurately.
-  Robots can work in hazardous environments without risk to humans.
-  Overall cost-saving over time (after setup).

[2]

(e) Give **two** disadvantages of using an automated robot in a factory.

Acceptable points (**Any two** for full marks):

-  High initial cost for purchase and installation.
-  Maintenance and repair costs can be expensive.
-  Can lead to job losses for human workers.
-  Robots may not handle unexpected situations as well as humans.
-  Programming and upgrading require specialized skills.
-  Downtime can occur if technical faults happen.

[2]

(f) The robot is upgraded to use machine learning.

Explain how this could improve the robot's performance.

Acceptable points (**Any two** for full marks):

-  The robot can learn from experience to improve accuracy over time.
-  It can adapt to new or changing environments without needing reprogramming.
-  It can detect and correct its own errors.
-  It can optimize routes or processes for faster or more efficient operation.
-  It can recognize patterns (e.g., identifying objects faster or more reliably).

[2]





Paper 1

9 A company owner has installed a new network. Data is correct before it is transmitted across the network.

The company owner is concerned that data might have errors after transmission.

(a) Explain how the data might have errors after transmission.

Acceptable points (**Any three** for full marks):

-  Electrical interference during transmission
-  Signal degradation over long distances
-  Faulty cables or connectors
-  Collisions on a network (especially in shared media networks)
-  Noise on the transmission medium
-  Incorrect routing or transmission protocols causing corruption

[3]

(b) The company owner decides to introduce an error detection system to check the data for errors after transmission.

The error detection system uses an odd parity check and a positive automatic repeat query (ARQ).

(i) Describe how the error detection system operates to check for errors.

Acceptable points (**Any eight** for full marks):

-  Sender calculates and adds a parity bit to each byte of data (odd parity).
-  Parity bit ensures the total number of 1's is odd.
-  Data and parity bit are transmitted to receiver.
-  Receiver checks the received data against odd parity rule.
-  If correct parity is detected, receiver sends positive acknowledgment (ACK).
-  If incorrect parity, receiver sends negative acknowledgment (NACK).
-  If a NACK or no response is received within a timeout, sender resends the data.
-  This repeats until correct data is acknowledged or transmission attempts fail after a number of retries.

[8]

(ii) Give **two** other error detection methods that could be used.

Acceptable examples:

-  Checksum
-  Cyclic Redundancy Check (CRC)
-  Echo check

[2]





Paper 1

(c) The company owner also installs a firewall to help protect the network from hackers and malware.

(i) Explain how the firewall operates to help protect the network.

Acceptable points (**Any five** for full marks):

-  Firewall monitors incoming and outgoing traffic.
-  It blocks unauthorized access based on security rules.
-  It can allow or block traffic based on IP addresses, protocols, ports.
-  It prevents access from known malicious addresses.
-  It may also inspect packets for suspicious content.
-  Alerts administrators of unauthorized access attempts.
-  Filters traffic based on predefined rules or dynamic analysis.

[5]

(ii) Give **two** examples of malware that the firewall can help protect the network from and how the malware work.

Acceptable examples (2 malware + how they work):

Malware: **How it works**

Virus: Attaches to files and spreads when the file is opened, damaging or corrupting data.

Worm: Self-replicates over the network without user action, consuming bandwidth and system resources.

Trojan horse: Disguises as legitimate software, then executes malicious activities once installed.

Spyware: Secretly gathers user information without permission.

[4]

